CONTROLLER / USB

ISOLATED POWER + I2C

ANALOG POWER / REFERENCE

J4
DISPLAY HARNESS
1 USB5V
2 USB_GND
3 HOST3V3
4 SDA_HOST
5 SCL_HOST

F1
250mA PTC
USB5V — USB5V_F

U4
TLV75533PDBVR
USB5V_F → IN
USB5V_F → EN
USB_GND → GND
OUT → P3V3
NC

U7
B0305S-1WR3
P3V3 → VIN
USB_GND → GND1
VOUT → ISO5V_RAW
GND2 → AGND

U5
TPS70933DBVR
ISO5V_RAW → IN
AGND → EN
GND
OUT → A3V3
NC

R37
4.7k
A3V3 — VBIAS

C16
10u
USB5V_F — USB_GND

C17
100n
USB5V_F — USB_GND

D1
LDO reverse protection
USB5V_F → P3V3

C20
10u
ISO5V_RAW — AGND

C21
100n
ISO5V_RAW — AGND

C22
10u
A3V3 — AGND

C23
100n
A3V3 — AGND

R38
1k
VBIAS — AGND

C26
470p
VBIAS — AGND

Waveshare ESP32-S3-LCD-1.28
J4: 1=5V, 2=GND, 3=3V3, 4=SDA/GPIO6, 5=SCL/GPIO7
R46/R47 DNP: host module already has 4.7k pull-ups.

C18
10u
P3V3 — USB_GND

C19
100n
P3V3 — USB_GND

R39
1k
ISO5V_RAW — AGND

R40
1k
ISO5V_RAW — AGND

R41
1k
ISO5V_RAW — AGND

R42
1k
ISO5V_RAW — AGND

R43
1k
ISO5V_RAW — AGND

U6
ISO1640BDR
HOST3V3 → VCC1
SDA_HOST → SDA1
SCL_HOST → SCL1
USB_GND → GND1
VCC2 → A3V3
SDA2 → SDA_ISO
SCL2 → SCL_ISO
GND2 → AGND

U8
ADS1115IDGSR
REF_ADC → AIN0
AGND → AIN1
AGND → AIN2
AGND → AIN3
SCL_ISO → ADDR
VDD → A3V3
GND → AGND
SDA → SDA_ISO
SCL → SCL_ISO
ALERT_RDY

R46
4.7k
HOST3V3 — SDA_HOST

R47
4.7k
HOST3V3 — SCL_HOST

C24
100n
HOST3V3 — USB_GND

C25
100n
A3V3 — AGND

R44
4.7k
A3V3 — SDA_ISO

R45
4.7k
A3V3 — SCL_ISO

R48
1k
A3V3 — AGND

C27
100n
A3V3 — AGND

TP13 USB5V
TP14 USB_GND
TP15 HOST3V3
TP16 P3V3
TP17 ISO5V_RAW
TP18 A3V3
TP19 AGND
TP20 VBIAS
TP21 SDA_ISO
TP22 SCL_ISO

USB5V
USB5V_F
USB_GND
HOST3V3

R49
4.7k
VBIAS — REF_ADC

C28
470p
REF_ADC — AGND

All DUT pins remain isolated from USB ground.
Functional isolation only: not a mains/CAT-rated instrument.

Inputs 1–4
File: bank1.kicad_sch

Inputs 5–8
File: bank2.kicad_sch

Inputs 9–12
File: bank3.kicad_sch

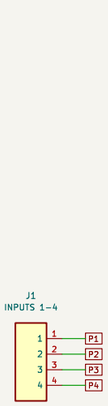
Prototype: bench validation required
12 V nominal DC | USB-powered measurement only
Power supply tester

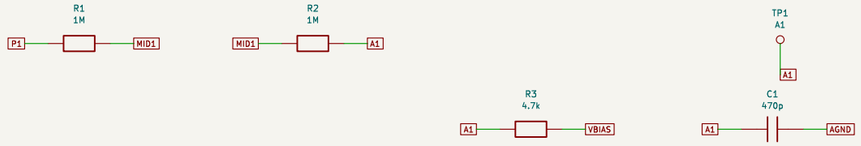
Sheet: /
File: tester12v.kicad_sch

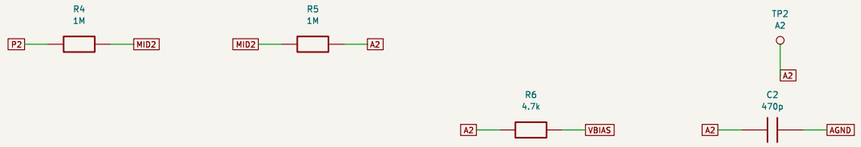
Title: 12 V multipin detector Rev D – isolated USB power

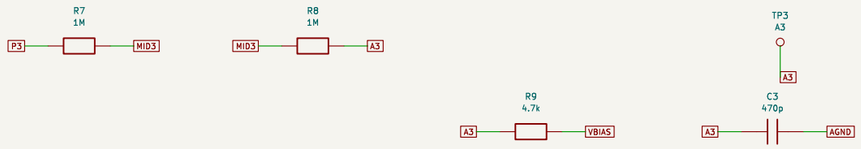
Size: A3	Date: 2026-09-30	**Rev: D**
KiCad E.D.A. 9.0.6		Id: 1/4

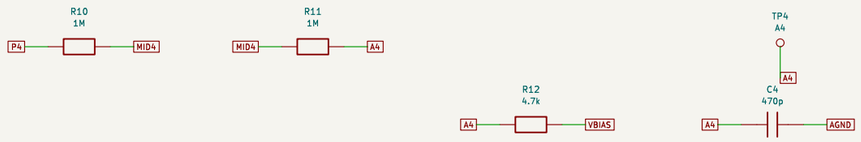
INPUTS 1–4: PASSIVE SENSING AND ADC

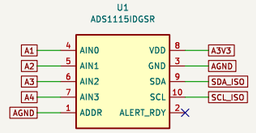














Two 1M resistors limit input current.
4.7k returns each ADC input to passive VBIAS.
No analog switch or active input clamp.
Input attenuation also applies without USB power.

12V nominal; 0–18V intended measurement range.
48V is a calculation stress, not a survival rating.
Do not connect any terminal directly to AGND or USB_GND.

470pF filters are on the ADC side.
Nominal attenuation = 426.5319. PGA = +/–1.024V.
One common supply; multiple + / return / open pins.

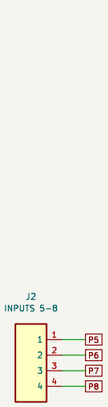
Prototype: bench validation required
12 V nominal DC | USB–powered measurement only
Power supply tester

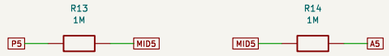
Sheet: /Inputs 1–4/
File: bank1.kicad_sch

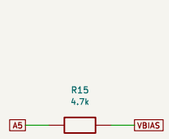
Title: Inputs 1–4: passive sensing and ADC

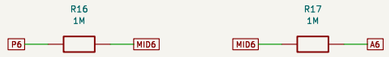
Size: A3	Date: 2026–09–30	**Rev: D**
KiCad E.D.A. 9.0.6		Id: 2/4

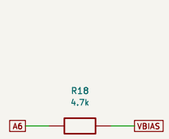
INPUTS 5–8: PASSIVE SENSING AND ADC

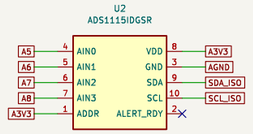


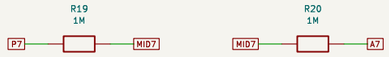


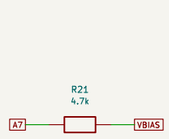






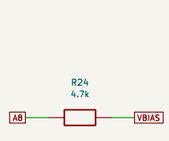












Two 1M resistors limit input current.
4.7k returns each ADC input to passive VBIAS.
No analog switch or active input clamp.
Input attenuation also applies without USB power.

12V nominal; 0–18V intended measurement range.
48V is a calculation stress, not a survival rating.
Do not connect any terminal directly to AGND or USB_GND.

470pF filters are on the ADC side.
Nominal attenuation = 426.5319. PGA = +/–1.024V.
One common supply; multiple + / return / open pins.

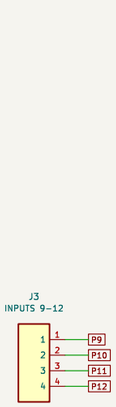
Prototype: bench validation required
12 V nominal DC | USB–powered measurement only
Power supply tester

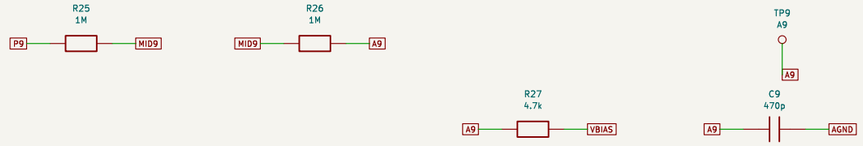
Sheet: /Inputs 5–8/
File: bank2.kicad_sch

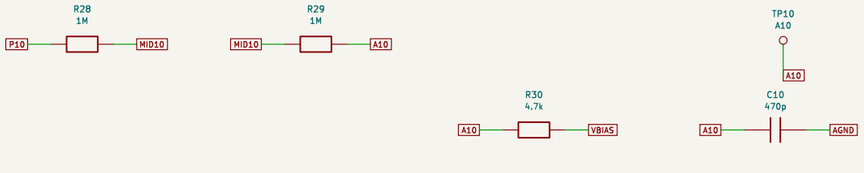
Title: Inputs 5–8: passive sensing and ADC

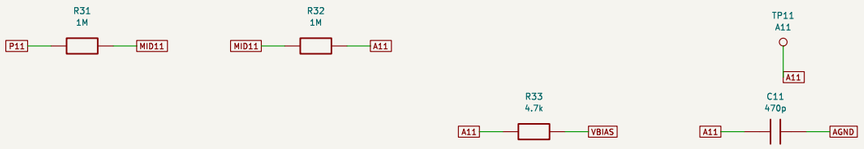
Size: A3	Date: 2026–09–30	**Rev: D**
KiCad E.D.A. 9.0.6		Id: 3/4

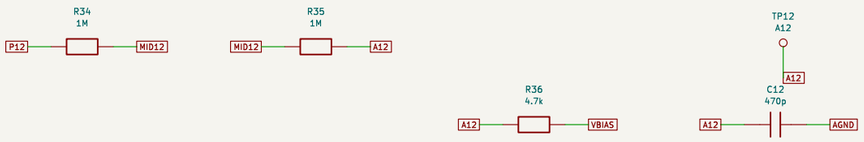
INPUTS 9–12: PASSIVE SENSING AND ADC

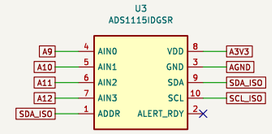














Two 1M resistors limit input current.
4.7k returns each ADC input to passive VBIAS.
No analog switch or active input clamp.
Input attenuation also applies without USB power.

12V nominal; 0–18V intended measurement range.
48V is a calculation stress, not a survival rating.
Do not connect any terminal directly to AGND or USB_GND.

470pF filters are on the ADC side.
Nominal attenuation = 426.5319. PGA = +/–1.024V.
One common supply; multiple + / return / open pins.

Prototype: bench validation required
12 V nominal DC | USB–powered measurement only
Power supply tester

Sheet: /Inputs 9–12/
File: bank3.kicad_sch

Title: Inputs 9–12: passive sensing and ADC

Size: A3	Date: 2026–09–30	**Rev: D**
KiCad E.D.A. 9.0.6		Id: 4/4